**sPHENIX Internal Note**

# Pileup Correction to Vertex + MBD Acceptance (Luminosity Note Section 3)

Isolated Photon Cross-Section in $p$+$p$ at $\sqrt{s} = 200$ GeV

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**Abstract**

This note quantifies the pileup correction to the vertex + MBD coincidence acceptance in the sPHENIX isolated photon cross-section measurement (PPG12). In double-interaction events, the overlay collision provides additional MBD hits and shifts the reconstructed vertex toward the beam center, increasing the effective acceptance from $\varepsilon_{\mathrm{vtx+MBD}} = 0.68$ (single interaction) to 0.81 (double interaction) — a 19% relative increase. After Poisson weighting with the measured double-interaction fractions, the net correction to the cross-section is $-2.1\%$ for 0 mrad ($f_{\mathrm{double}} = 11.1\%$) and $-0.7\%$ for 1.5 mrad ($f_{\mathrm{double}} = 3.9\%$), with a mild $p_{\mathrm{T}}$-dependence rising from $+2\%$ at 8 GeV to $+3.5\%$ at 30 GeV. This correction is currently not applied in the production efficiency chain, which uses single-interaction MC only.

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# 1 Motivation

The sPHENIX luminosity note (Section 3) describes a correction needed when the analysis requires both an MBD coincidence and a z-vertex selection. In PPG12, events must satisfy:

- MBD N&S $\geq 1$ (at least one hit on each side)
- $|z_{\rm vtx}^{\rm reco}| < 60$ cm

The production efficiency chain (`RecoEffCalculator_TTreeReader.C` → `CalculatePhotonYield.C`) computes $\varepsilon_{\rm vtx+MBD}$ from single-interaction MC:

$$\varepsilon_{\rm vtx+MBD} = \frac{N(\text{truth photons passing MBD + vertex cut})}{N(\text{truth photons with } |z_{\rm vtx}^{\rm truth}| < 60 \text{ cm})} \tag{1}$$

In data, pileup events modify this acceptance in two ways:

1. **MBD rescue:** If the signal collision fires only one MBD side (or neither), the overlay collision can provide the missing hits, satisfying the coincidence requirement.
2. **Vertex centering:** The MBD-reconstructed vertex is approximately the average of two collision vertices. This pulls the vertex toward zero, increasing the fraction passing the $|z_{\rm vtx}| < 60$ cm cut.

The correction is:

$$\varepsilon_{\rm vtx+MBD}^{\rm blended} = (1 - f_{\rm double}) \times {\rm Pass}_1 + f_{\rm double} \times {\rm Pass}_2 \tag{2}$$

where $\text{Pass}_1$ is the single-interaction acceptance and $\text{Pass}_2$ is the double-interaction acceptance, and $f_{\rm double}$ is the event-level double-interaction fraction from Poisson statistics.

# 2 Method

## 2.1 Samples

Single-interaction MC uses the three standard photon-jet samples, weighted by production cross-section:

- `photon5` (0–14 GeV), weight $= \sigma_5/\sigma_{20} = 1122$
- `photon10` (14–30 GeV), weight $= \sigma_{10}/\sigma_{20} = 53.2$
- `photon20` (30+ GeV), weight $= 1.0$ (reference)

Double-interaction MC uses the full GEANT4 double-interaction sample:

- `photon10_double` — two independent pp collisions simulated per event, with the photon from the primary collision in the 10–100 GeV truth $p_{\rm T}$ range.

Two million events are processed per sample.

## 2.2 Truth photon selection

Standard PPG12 truth criteria: `pid` $= 22$, `photonclass` $< 3$, $E_{\rm T}^{\rm iso,truth} < 4$ GeV (cone $R = 0.3$), $|\eta| < 0.7$, $8 \leq p_{\rm T} < 36$ GeV.

The denominator of Eq. (1) requires $|z_{\rm vtx}^{\rm truth}| < 60$ cm (same as `RecoEffCalculator_TTreeReader.C:1694`).

### 2.3 Acceptance categories

For each truth photon in the denominator, we check four acceptance conditions independently:

1. **MBD + vertex (combined):** MBD N&S $\geq 1$ AND $|z_{\rm vtx}^{\rm reco}| < 60$ cm — this is $\varepsilon_{\rm vtx+MBD}$ used in the cross-section formula.
2. **MBD only:** MBD N&S $\geq 1$ (no reco vertex cut).
3. **Vertex only:** $|z_{\rm vtx}^{\rm reco}| < 60$ cm (no MBD requirement).
4. **MBD hit decomposition:** classified into both N&S, N only, S only, neither.

### 2.4 Pileup fractions

Event-level double-interaction fractions from `calc_pileup_range.C` (Poisson statistics, lumi-weighted):

- 0 mrad: $f_{\rm double} = 11.1\%$ (event-level; 22.4% cluster-weighted)
- 1.5 mrad: $f_{\rm double} = 3.9\%$ (event-level; 7.9% cluster-weighted)

Triple+ interactions are folded into the double fraction ($\text{Prob}_3 < 0.3\%$ for $\mu < 0.3$).

## 3 Results

### 3.1 Single vs. double-interaction acceptance

Figure 1 shows the per-$p_{\rm T}$ vertex+MBD acceptance for single-interaction (blue) and double-interaction (red) MC. The double-interaction acceptance is systematically higher across all $p_{\rm T}$ bins, with a ratio of $\sim$1.19 (Figure 2).

### 3.2 Acceptance breakdown

Figure 3 separates the MBD and vertex contributions:

- **MBD acceptance:** increases from 69% (single) to 92% (double) — a 33% relative increase. The overlay collision rescues events where the signal fired only one MBD side or neither.
- **Vertex acceptance:** increases from 75% (single) to 83% (double) — an 11% relative increase. The averaged vertex is narrower than individual vertices.

The MBD rescue effect dominates the correction.

### 3.3 MBD hit decomposition

Figure 4 shows the stacked MBD hit fractions. In single-interaction MC, $\sim$8% of truth photons produce no MBD hits ("neither"), consistent with the GEANT4 prediction ($P_{\rm neither} = 8.8\%$). In double-interaction MC, this drops to $< 1\%$ because the overlay almost always provides some MBD activity.

### 3.4 Correction factors

Figure 5 shows the per-$p_{\rm T}$ correction factor $\varepsilon_{\rm vtx+MBD}^{\rm blended}/\varepsilon_{\rm vtx+MBD}^{\rm single}$ for both crossing-angle periods.

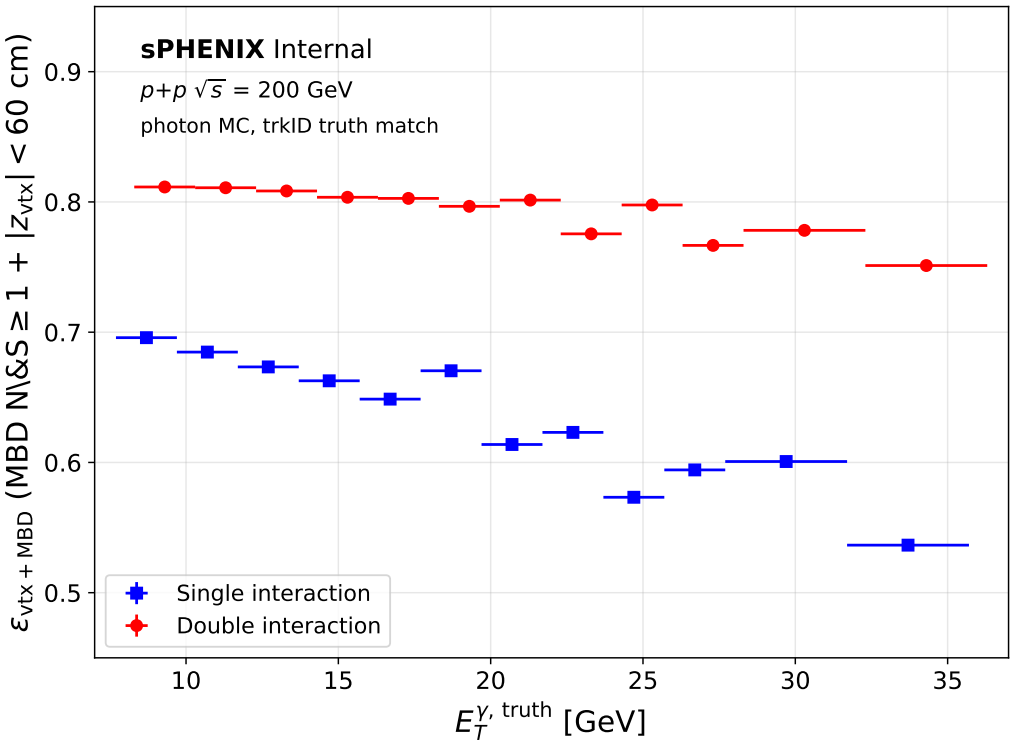


Figure 1: Per-$p_{\mathrm{T}}$ vertex+MBD acceptance for single-interaction (blue) and double-interaction (red) photon MC. Error bars are binomial. The double-interaction acceptance is 19% higher on average due to pileup MBD rescue and vertex centering.

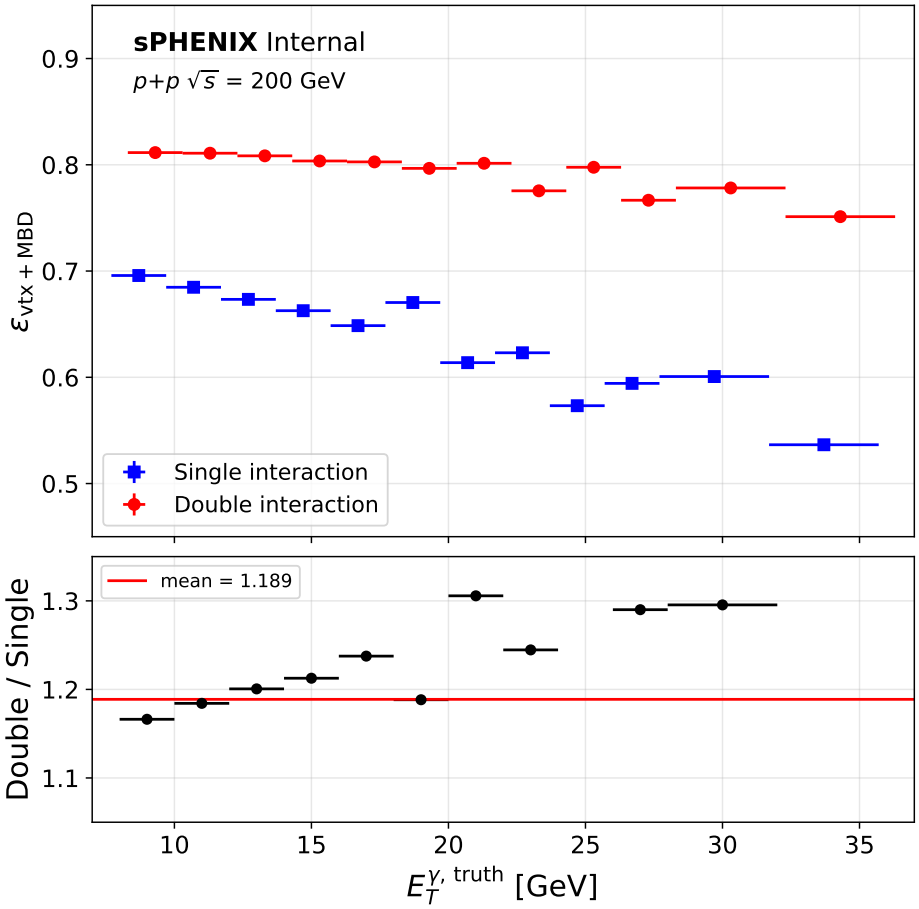


Figure 2: Top: vertex+MBD acceptance for both interaction types. Bottom: ratio (double/single), showing a ∼1.19 enhancement that increases slightly with $p_{\mathrm{T}}$ (from 1.17 at 8 GeV to 1.28 at 30 GeV).

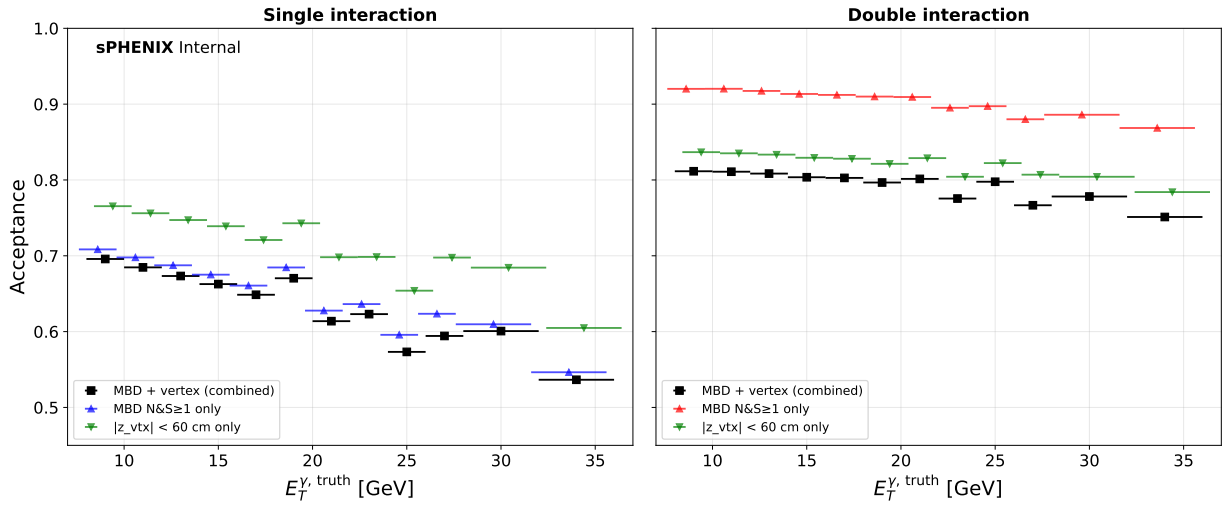


Figure 3: Acceptance breakdown into MBD-only (triangles up), vertex-only (triangles down), and combined (squares) for single (left) and double (right) interaction MC.

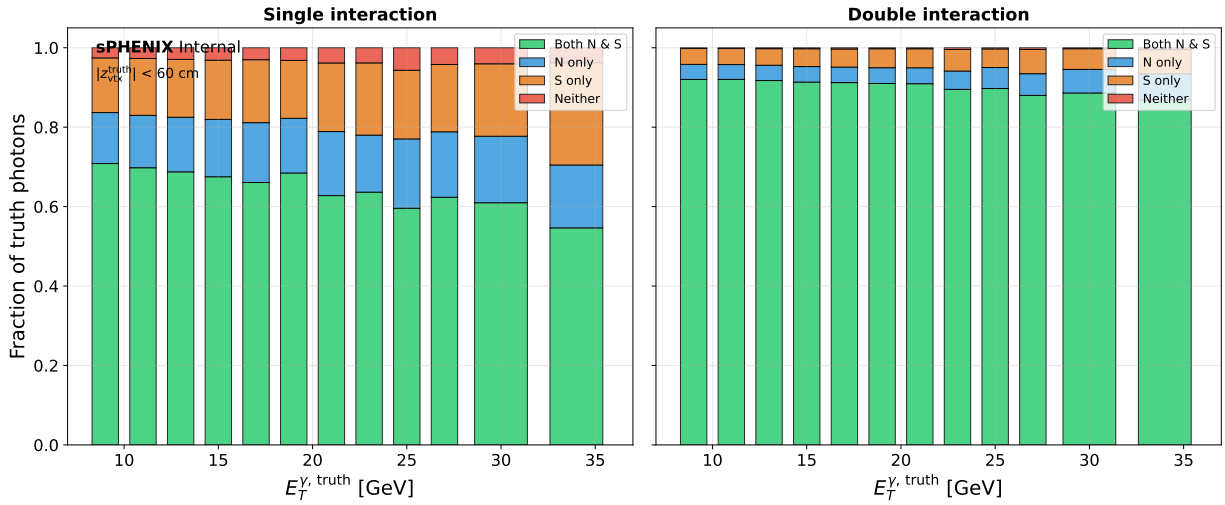


Figure 4: MBD hit decomposition for truth photons within the vertex acceptance. "Both N&S" = MBD coincidence satisfied; "N only" / "S only" = single-arm fire; "Neither" = no MBD hits. The double-interaction overlay nearly eliminates the "neither" category.

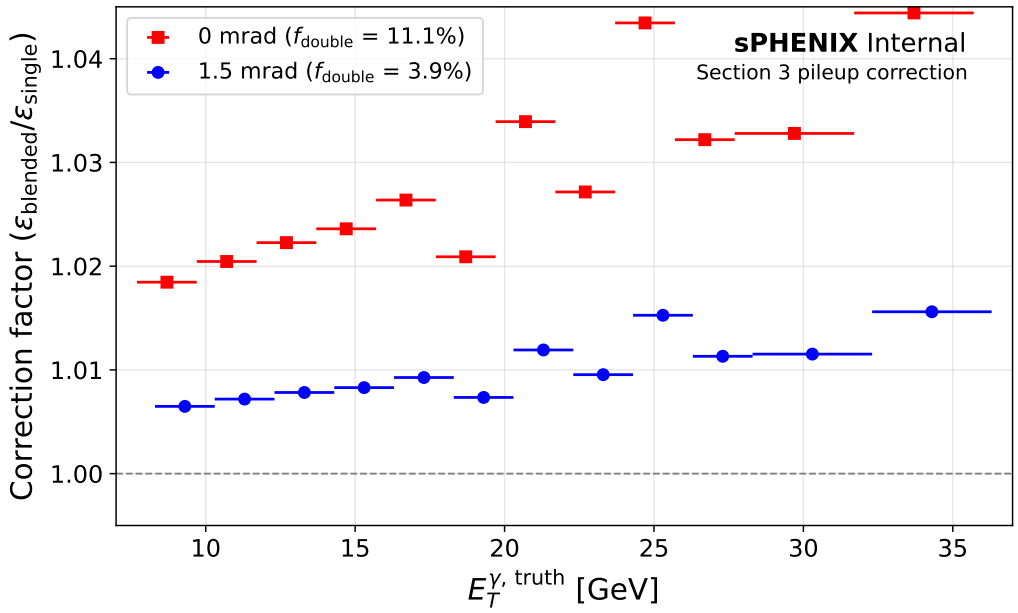


Figure 5: Per-$p_{\mathrm{T}}$ correction factor for 0 mrad (red, $f_{\text{double}} = 11.1\%$) and 1.5 mrad (blue, $f_{\text{double}} = 3.9\%$). The correction increases with $p_{\mathrm{T}}$ because the single-interaction acceptance drops faster at high $p_{\mathrm{T}}$ (forward photons more sensitive to vertex shifts), while double-interaction rescues proportionally more events.

# 4 Impact on the Cross-Section

The correction increases $\varepsilon_{\text{vtx+MBD}}$ in the efficiency denominator, which *decreases* the measured cross-section:

$$\left.\frac{d\sigma}{dp_{\mathrm{T}}}\right|_{\text{corrected}} = \left.\frac{d\sigma}{dp_{\mathrm{T}}}\right|_{\text{current}} \times \frac{\varepsilon^{\text{single}}_{\text{vtx+MBD}}}{\varepsilon^{\text{blended}}_{\text{vtx+MBD}}} \tag{3}$$

Table 1: Integrated correction factors and cross-section impact.

| **Period** | $f_{\text{double}}$ | **Integrated correction** | **Cross-section change** |
|---|---|---|---|
| 0 mrad | 11.1% | +2.1% | −2.1% |
| 1.5 mrad | 3.9% | +0.7% | −0.7% |

The $p_{\mathrm{T}}$-dependent correction ranges from +2.0% at 8 GeV to +3.7% at 26 GeV for the 0 mrad period.

# 5 Discussion

## 5.1 Comparison with other systematics

The Section 3 correction is smaller than the dominant PPG12 systematics:

- Purity systematic: 5–10%

- Energy scale: 3–5%
- $\sigma_{\mathrm{MBD}}$ uncertainty: $\sim$7% (flat)

For the 1.5 mrad nominal result, the +0.7% correction is negligible. For 0 mrad or combined, the +2.1% (rising to 3.5% at high $p_{\mathrm{T}}$) is comparable to the energy scale systematic and should be applied or assigned.

### 5.2 Limitations

1. Only `photon10_double` exists as full GEANT double-interaction MC. The correction uses the same $\mathrm{Pass}_2$ for all $p_{\mathrm{T}}$, though the actual correction should be computed per truth $p_{\mathrm{T}}$ window (photon5/10/20).
2. Triple+ interactions ($n \geq 3$) are folded into the double fraction. At $\mu \approx 0.12$ (0 mrad), $\mathrm{Prob}_3 \approx 0.3\%$ — negligible.
3. The correction is computed as a single average over all runs, not per-segment. The $\mu$ variation within a fill introduces a $< 0.5\%$ residual uncertainty (Jensen's inequality).
4. The double-interaction vertex distribution is from GEANT4 simulation; any data-MC discrepancy in the beam vertex profile propagates.

## 6 Recommendation

1. **1.5 mrad (nominal):** No correction needed. Assign +0.7% as a one-sided systematic if desired.
2. **0 mrad or combined:** Apply the per-$p_{\mathrm{T}}$ correction from Table in Figure 5, or assign +2.1% (integrated) as a systematic. The $p_{\mathrm{T}}$-dependent correction table is available in `results/pileup_vertex_acce`
3. **Implementation:** Multiply the current `vertex_eff` in `CalculatePhotonYield.C:1009` by the per-$p_{\mathrm{T}}$ correction factor for the appropriate crossing-angle period.

## Files

| File | Purpose |
|---|---|
| `efficiencytool/study_pileup_vertex_acceptance.py` | Analysis script |
| `plotting/plot_pileup_vertex_correction.py` | Plotting script |
| `results/pileup_vertex_acceptance.root` | ROOT output |
| `results/pileup_vertex_acceptance.csv` | Per-$p_{\mathrm{T}}$ table |
| `plotting/figures/pileup_vtx/*.pdf` | Figures |